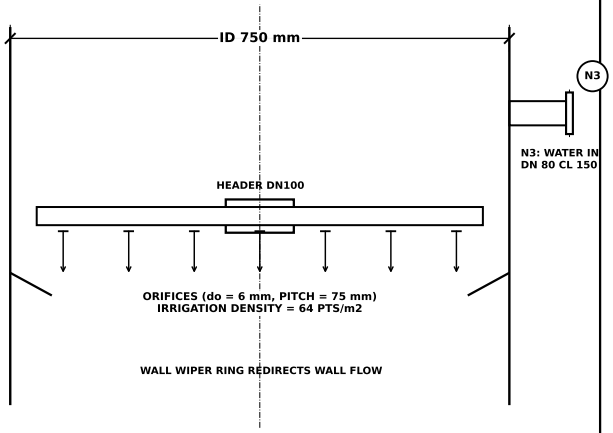
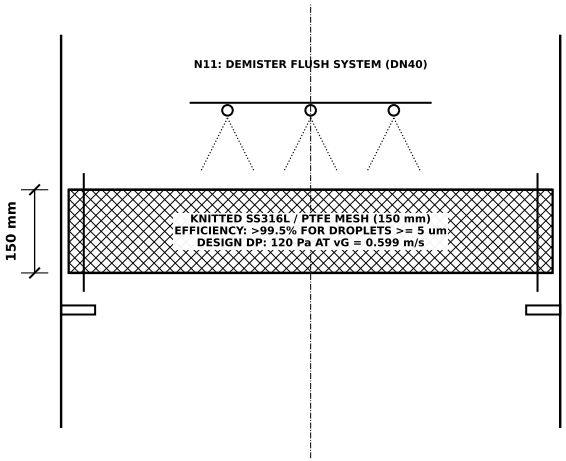
DETAIL A — LIQUID DISTRIBUTOR

HEADER-LATERAL MATRIX (SCALE 1:20 METRIC)



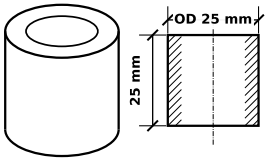
DETAIL B — MIST ELIMINATOR (DEMISTER)

KNITTED WIRE MESH (150 mm)— SCALE 1:20 METRIC



DETAIL C — 25-mm CERAMIC RASCHIG RING

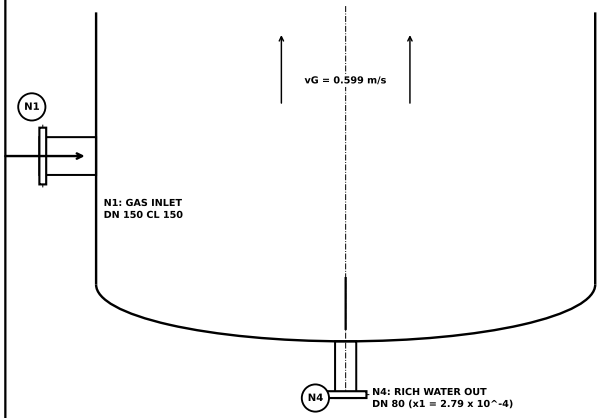
ORTHOGRAPHIC & 3D ISOMETRIC (SCALE 1:2 METRIC)



CERAMIC PACKING SPECIFICATION:
- PACKING RANDOM-DUMPED; NOMINAL SIZE 25 mm
- PACKING BED DEPTH: Z = 4500 mm (Vp = 1.79 m3)
- Glazed Chemical Porcelain (ASTM C515)
- Specific Surface Area: a = 190 m2/m3
- Bed Voidage: eps = 0.74 (74% Free Void)
- Bulk Density: rho_bulk = 680 kg/m3
- Packing Factor: Fp = 100 m-1 (30 ft-1)
- Dry Packing Weight: 11.9 kN (1.22 t)

DETAIL D — GAS INLET & DEFLECTOR

AERODYNAMIC CHEVRON & SUMP (SCALE 1:20 METRIC)



INTERNALS SPECIFICATION SUMMARY

INTERNAL	MATERIAL / TYPE	FUNCTION / CAPACITY
Distributor	SS316L Header-Lateral	64 pts/m2 irrigation density
Redistributor	Titanium Wall Wiper	Captures wall-flow liquid
Hold-Down	SS316L Heavy Grid	Prevents ring fluidization
Packing Bed	Ceramic Raschig 25 mm	**Z = 4500 mm, Vp = 1.79 m3**
Support Grid	High-Free Area FRP/Ti	Free area >85% (anti-flood)
Gas Deflector	Titanium 60 DEG Chevron	Reduces 14 m/s to 0.6 m/s
Mist Demister	Knitted SS316L/PTFE	>99.5% droplet separation

INTERNALS BILL OF MATERIALS (BOM)

ITEM	DESCRIPTION	QTY	WEIGHT
1	Mist Demister Pad	1 SET	18.5 kg
2	Distributor Manifold	1 SET	45.0 kg
3	Wall Wiper Rings	2 PCS	14.8 kg
4	Ceramic Raschig Rings	1.79 m3	1220 kg
5	Support Grid & Cleats	1 SET	65.0 kg
6	Inlet Deflector Cone	1 PC	16.5 kg
7	Anti-Vortex Sump Baffle	1 SET	11.0 kg

FABRICATION & INSTALLATION NOTES

1. **DISTRIBUTOR: HORIZONTAL LEVEL +/-1.0 mm OVER 750 mm DIA.**
2. **WET PACKING: 25-mm RINGS DUMPED IN WATER-FILLED COLUMN.**
3. HIGH FREE AREA (>85%): MANDATORY ON PACKING SUPPORT GRID.
4. FASTENERS: TITANIUM GR 2 / HASTELLOY C-276 IN WETTED ZONES.

JADAVPUR UNIVERSITY | CHEMICAL ENGINEERING DEPARTMENT | PROCESS EQUIPMENT DESIGN & DRAWING

DRAWING TITLE	LAB / SUBJECT	COURSE / BRANCH	SCALE
PACKED-BED ABSORBER (C-101) — INTERNAL ASSEMBLY CAD	PROCESS EQUIPMENT DESIGN & DRAWING	B.E. CHEMICAL ENGINEERING	DETAILED 1:2, 1:20
STUDENT NAME	**ROLL NO.**	**DRAWING NO.**	**DATE**
SOURISH SENAPATI	002310301035	PEDD-PBA-02	29 / 09 / 2026

PACKED ABSORBER (C-101) | 0.25 m3/s GAS (2% Br2) | WATER: 17.25 kg/s | ASME SEC VIII DIV 1 / B16.5